DealerData Intelligence

Automotive Scraper Platform Support & Architecture Specification

October 2026
Scraper Engine v2.4 (Enterprise)

EXECUTIVE ARCHITECTURE SUMMARY

The DealerData scraping subsystem implements a dual-architecture paradigm: **dedicated native adapters** optimized for top tier automotive CMS vendors (Dealer.com, Dealer Inspire, DealerOn, CDK Global, etc.) coupled with an autonomous **Universal Cascading Funnel** capable of crawling any dealer website via XML sitemaps, Schema.org JSON-LD extraction, and headless stealth Playwright automation.

01 Dedicated Automotive Platform Adapters

Platforms with dedicated drivers, internal state extractors, and direct REST/GraphQL feeds:

PLATFORM / CODE	VENDOR & ECOSYSTEM	EXTRACTION MECHANISM	KEY FEATURES & CAPABILITIES
Dealer.com `dealercom`	Cox Automotive	Direct SRP + `DDC.WS.state` + Playwright	• Parses internal state `DDC.WS.state['ws-inv-data']` • Native WIS incentive, rebate & lease parser • DDC URL facet handler (`/new-inventory/index.htm`)
Dealer Inspire `dealerinspire`	Cars Commerce / Cars.com	CarsCommerce REST API + JSON-LD	• Direct API with `x-api-key` authorization • Fallback endpoint `/inventory/search?format=json` • High-res CDN image stream (`carscommerce.inc`)
DealerOn `dealeron`	DealerOn Inc.	JSON-LD Schema + SRP Card Engine	• Schema.org `ItemList` / `Vehicle` extraction • Native widget summary query probe • Clean VIN-keyed pricing & stock resolution
CDK Global `cdk`	CDK / DealerTrack	Runtime DOM + Playwright Stealth	• Detects `window.cdk` & vendor scripts • Handles complex multi-store dealer portals • Stealth anti-bot challenge bypass
Team Velocity `teamvelocity`	Apollo Automotive Sites	HTML SRP + Specials Selectors	• Apollo-specific inventory card & title parser • Dynamic pricing and incentive tags
Pixel Motion `pixelmotion`	Pixel Motion Automotive	HTML SRP + Media CDN Engine	• Custom layout facet filters & inventory tables • Banner specials via `pixelmotion.com` CDN
AutoFi `autofi`	AutoFi Inc.	Widget Probe + Dynamic DOM	• Digital retailing pricing overlay extraction • Payment breakdown (lease vs finance terms)

02 The Universal 4-Stage Cascading Funnel

For unlisted or custom dealer CMS websites (`mode: "auto"`), the crawler executes this sequential fallback funnel:

1 Sitemap Discovery & VDP Crawl

Probes `robots.txt` and sitemap indexes (`sitemap_inventory.xml`, `vehicles.xml`). Concurrently crawls direct vehicle URLs (VDPs) with zero browser overhead.

2 Direct Platform API & Feeds

Probes standard automotive REST endpoints and JSON feeds (CarsCommerce, `/api/inventory`, `WSInventorySummary`). Retrieves clean structured records instantly.

3 Direct Target URL SRP Crawl

Fetches search results pages directly using HTTP stealth headers. Parses embedded JSON-LD graphs and extracts cards across 11 universal automotive CSS selectors.

4 Stealth Playwright Headless Browser

Automates full Chromium browser with anti-detection evasions (pre-warmed origins, Cloudflare Turnstile handling, lazy-load auto-scroll, and DOM capture).

03 Specials, Lease & Visual Flyer Extraction Engine

Dealership specials are published either as structured HTML components or visual promotional flyer images. The scraper pipeline handles both seamlessly:

1. Structured HTML Component Specials

Extracts lease/finance parameters directly from HTML DOM or JSON-LD:

- **Monthly Payment:** Regex patterns for `$XX/mo`, `per month`
- **Lease Term:** Detects terms (e.g., 24, 36, 39, 48 months)
- **Down Payment (DAS):** Due-at-signing & down payment extraction
- **APR & Discounts:** Special APR rates & MSRP rebates
- Directly synced to `competitor_vehicle_specials`.

2. Visual Graphic Flyer Staging (AI Vision Ready)

When specials are embedded inside graphic flyers (e.g. J Star CDJR):

- **Smart CDN Filter:** Filters out logos, social icons, and tracking pixels
- **SHA-256 Content Hashing:** Computes `image_hash` on download
- **Global AI Cache:** Checks if image was parsed previously by AI in any workspace, avoiding duplicate Vision AI API costs
- Staged in `specials_staging`.

04 Current Production Store Distribution

Current distribution of active stores and competitors monitored in the database:

PLATFORM IDENTIFIER	CONFIGURATION MODE	STORES	COMPETITORS	TOTAL
Auto / Universal (`auto`)	Autonomous 4-Tier Funnel	3	5	**8**
Dealer Inspire (`dealerinspire`)	Native CarsCommerce API + JSON-LD	0	6	**6**
Dealer.com (`dealercom`)	DDC State + SRP Crawl	4	3	**7**
DealerOn (`dealeron`)	Schema.org ItemList + Search Widget	0	2	**2**
AutoFi (`autofi`)	AutoFi Retail Widget Probe	0	2	**2**
Team Velocity (`teamvelocity`)	Apollo Card Selectors	0	1	**1**

05 Efficiency, Caching & Anti-Block Safeguards

FEATURE	CONFIGURATION	OPERATIONAL BENEFIT
Global Domain Scrape Cache	`Strict 6-Hour TTL`	Prevents redundant browser launches when multiple stores monitor the same competitor domain. Instant sub-second response.
SHA-256 Flyer AI Deduplication	`image_hash`	Identical promotional flyers reuse completed AI Vision extractions globally, saving up to 90% of LLM processing fees.
Inherit-and-Override Store Overlay	`workspace_special_overrides`	Stores can customize or adjust canonical specials without breaking continuous sync with fresh global scrapes.
Stealth Cloudflare Resolution	`Pre-Warm + Turnstile`	Automated pre-warming via origin homepage eliminates 403 blocks on sensitive dealership CDNs.